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Lab 7.1 Database & Visulization 2 - Exploring Grafana Dashboard

Learning Goals

Students will be able to:

1. Extract information from a database using the Grafana Dashboard
2. Discuss the relationship between the MTConnect agent and Grafana
3. Create visualizations in the Grafana Dashboard by combining Grafana features and SQL statements
4. Build a monitoring system in Grafana that connects all of the sensors, MTConnect Adapaters, Agents, SQL database, and Grafana dashboard features.

1.1 Introduction

The topic of Lab7 is visualization using Grafana web dashboard. We will practice creating Grafana dashboard using the existing database first and then build up the entire system yourself to monitor sensor data using Grafana in real-time remotely. At the end of this lab, you will be able to create the entire real-time remote IIoT monitoring system from the sensors or IoT devices to a web-based dashboard.

1.2 Log-in Grafana web server

Let's start with log in the Grafana server. The Grafana server domain and the port number information for this lab are as follows.

- DNS: mepotrb16.ecn.purdue.edu
- Port: 3000
  - Please note that default port number of Grafana is 3000.
- Grafana web page URL: http://mepotrb16.ecn.purdue.edu:3000/

When you access the Grafana web page using a web browser on your laptop, you will see the log-in page as Figure 1. TA already made an account for you. The username is the same as the MySQL account name, i.e., your username for Grafana is 'firstnamelastname' based on Brightspace, e.g., John Doe's username for ME597 Grafana server is 'johndoe'. Or you can use your **Purdue email address** as the log-in information. And TA will let you know the password of your account.

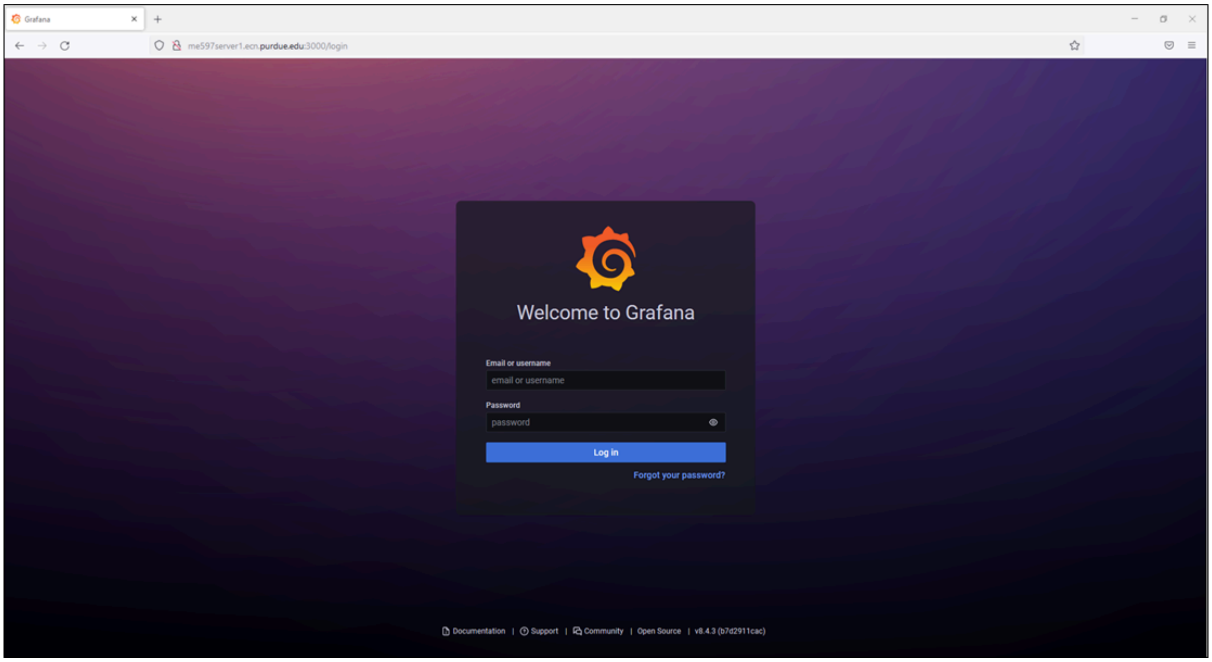


Figure 1 Log-in page of Grafana server

After successfully logging on to the Grafana server, you will see the main page as Figure 2. When you put the mouse pointer on the profile icon, and then you will see the profile menu. If you want to change password or information of your account, try it.

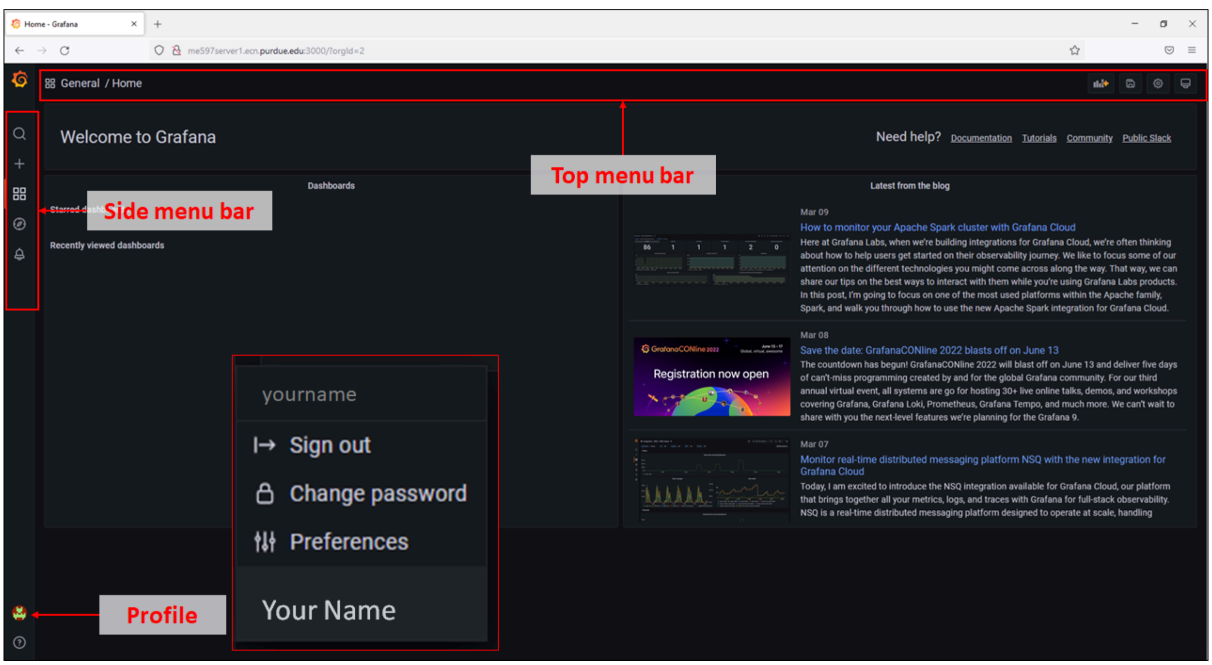


Figure 2 Grafana main page

1.3 Access To Dashboard

Before accessing the dashboard, let's understand the system and the data first. The schematic of the example system for the dashboards is shown in Figure 4. This is important because you should create the same system by accessing the data from sensors at the end of this lab. As we performed in Lab4, MTConnect adapter simulator is being used in this part. The target sensors are 1) DS18B20 & Virtual sensor, 2) ADXL345, and 3) power meter. The example data are shown in Figure 3. These example data for approximately 3 minutes are repeating continuously and infinitely in the MTConnect adapter simulator. All simulated data is transmitted to MTConnect agent.

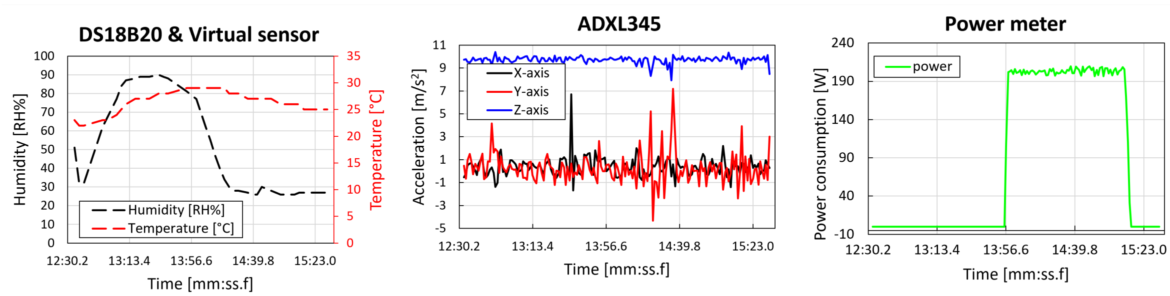


Figure 3 Example data for the example dashboard

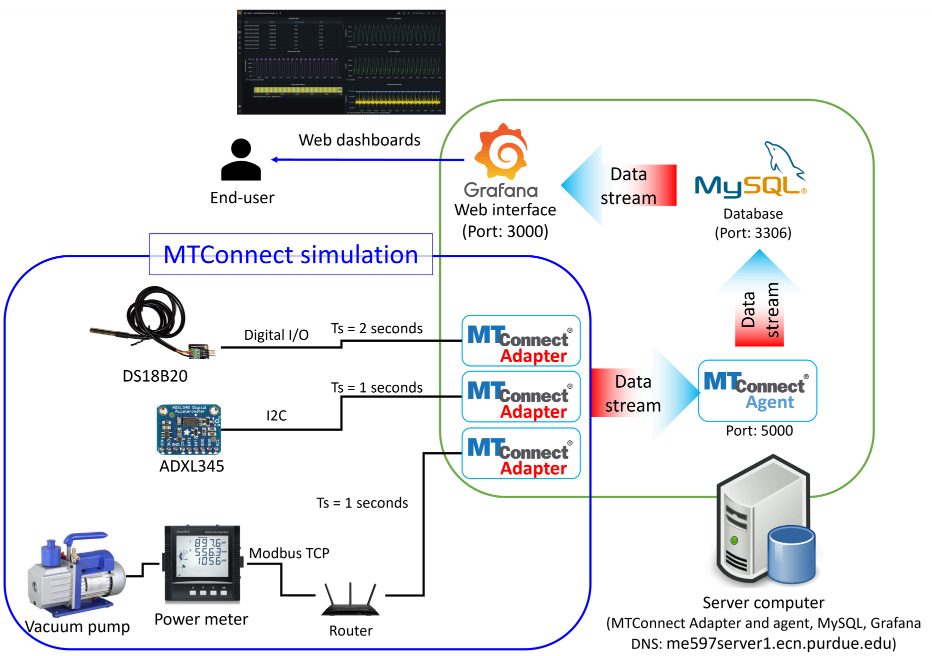
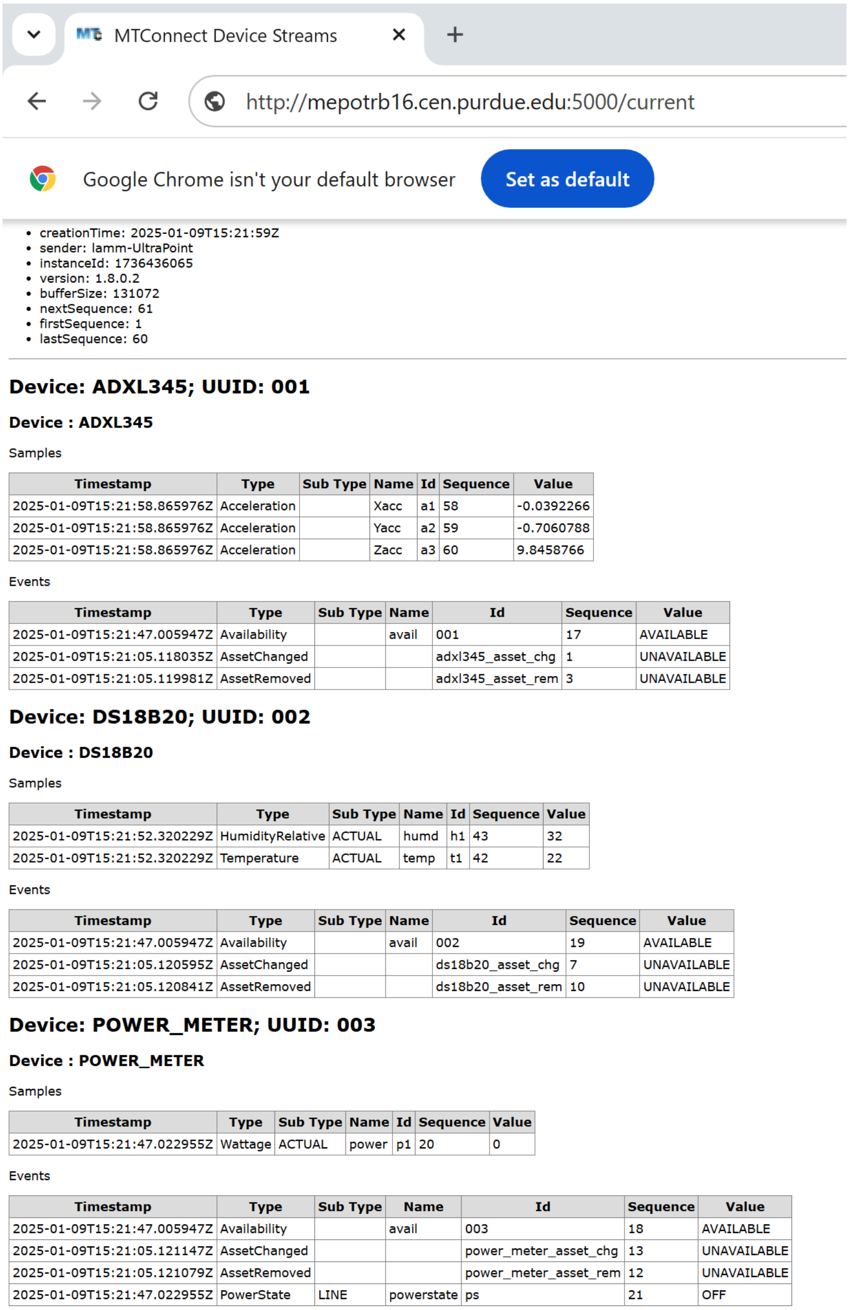


Figure 4 Schematic of example system

The information about the system is summarized as follows.

- MTConnect agent: http://mepotrb16.ecn.purdue.edu:5000/
- MySQL server DNS (same as Lab6): mepotrb16.ecn.purdue.edu
  - Port: 3306
  - Database: *
  - Table: grafana_sample
  - Table schema is the same as Lab6. But the only difference is the precision of the TIMESTAMP column. This is explained in the later section.
- Sensor
  - DS18B20
    - Temperature (Sample) in °C unit
  - Virtual sensor
    - Humidity (Sample) in %RH unit
  - ADXL345
    - X-axis acceleration (Sample) in m/s2 unit
    - Y-axis acceleration (Sample) in m/s2 unit
    - Z-axis acceleration (Sample) in m/s2 unit
  - Power meter
    - True power (Sample) of vacuum pump in Watt unit
    - Power state (Event): 'ON' or 'OFF'
      - When the true power is larger than 30 W, the result is 'ON' which means the vacuum pump is running.

If you request current information to MTConnect agent, you will see the result as Figure 5 (This window is captured on the server PC. You don't need to do this on your PC.). By aggregating MTConnect agent data, the MySQL database ('grafana_sample' table) is updated continuously. **If nothing is shown in your browser, please close the "Pretty" style following the instruction of Task 2.1 in L4 Colab 2.**



MTConnect Device Streams

http://mepotrb16.cen.purdue.edu:5000/current

Google Chrome isn't your default browser Set as default

- creationTime: 2025-01-09T15:21:59Z
- sender: lamm-UltraPoint
- instanceId: 1736436065
- version: 1.8.0.2
- bufferSize: 131072
- nextSequence: 61
- firstSequence: 1
- lastSequence: 60

Device: ADXL345; UUID: 001

Device : ADXL345

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2025-01-09T15:21:58.865976Z	Acceleration		Xacc	a1	58	-0.0392266
2025-01-09T15:21:58.865976Z	Acceleration		Yacc	a2	59	-0.7060788
2025-01-09T15:21:58.865976Z	Acceleration		Zacc	a3	60	9.8458766

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2025-01-09T15:21:47.005947Z	Availability		avail	001	17	AVAILABLE
2025-01-09T15:21:05.118035Z	AssetChanged			adxl345_asset_chg	1	UNAVAILABLE
2025-01-09T15:21:05.119981Z	AssetRemoved			adxl345_asset_rem	3	UNAVAILABLE

Device: DS18B20; UUID: 002

Device : DS18B20

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2025-01-09T15:21:52.320229Z	HumidityRelative	ACTUAL	humd	h1	43	32
2025-01-09T15:21:52.320229Z	Temperature	ACTUAL	temp	t1	42	22

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2025-01-09T15:21:47.005947Z	Availability		avail	002	19	AVAILABLE
2025-01-09T15:21:05.120595Z	AssetChanged			ds18b20_asset_chg	7	UNAVAILABLE
2025-01-09T15:21:05.120841Z	AssetRemoved			ds18b20_asset_rem	10	UNAVAILABLE

Device: POWER_METER; UUID: 003

Device : POWER_METER

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2025-01-09T15:21:47.022955Z	Wattage	ACTUAL	power	p1	20	0

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2025-01-09T15:21:47.005947Z	Availability		avail	003	18	AVAILABLE
2025-01-09T15:21:05.121147Z	AssetChanged			power_meter_asset_chg	13	UNAVAILABLE
2025-01-09T15:21:05.121079Z	AssetRemoved			power_meter_asset_rem	12	UNAVAILABLE
2025-01-09T15:21:47.022955Z	PowerState	LINE	powerstate	ps	21	OFF

Figure 5 Capture of current response from MTConnect agent

If you check the data in 'grafana_sample' table using SELECT statement as below on MySQL Workbench, you will see the result grid as Figure 6. Try this on MySQL Workbench yourself. If you compare the MTConnect agent data and MySQL database, you can confirm that all data in the MTConnect are aggregated into the database in real-time.

SQL - SELECT statement to see the table in descending order by timestamp

```sql
SELECT * FROM ME597Spring25.grafana_sample
ORDER BY timestamp DESC
```

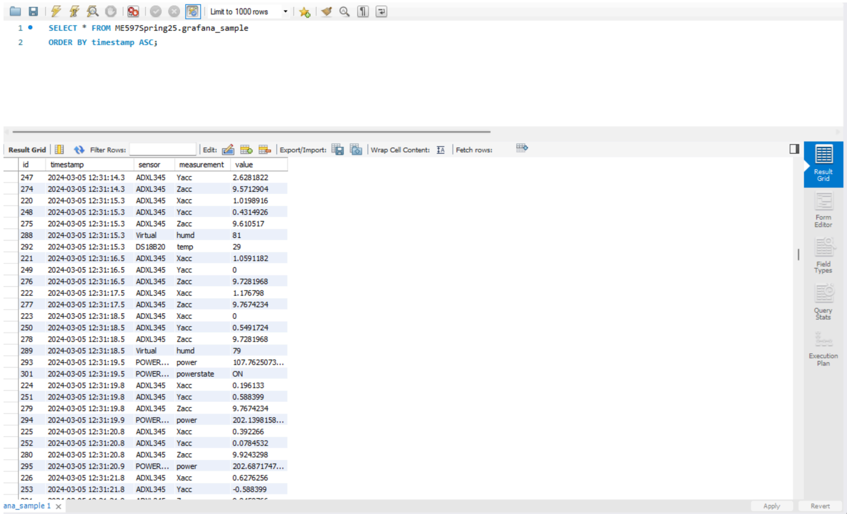
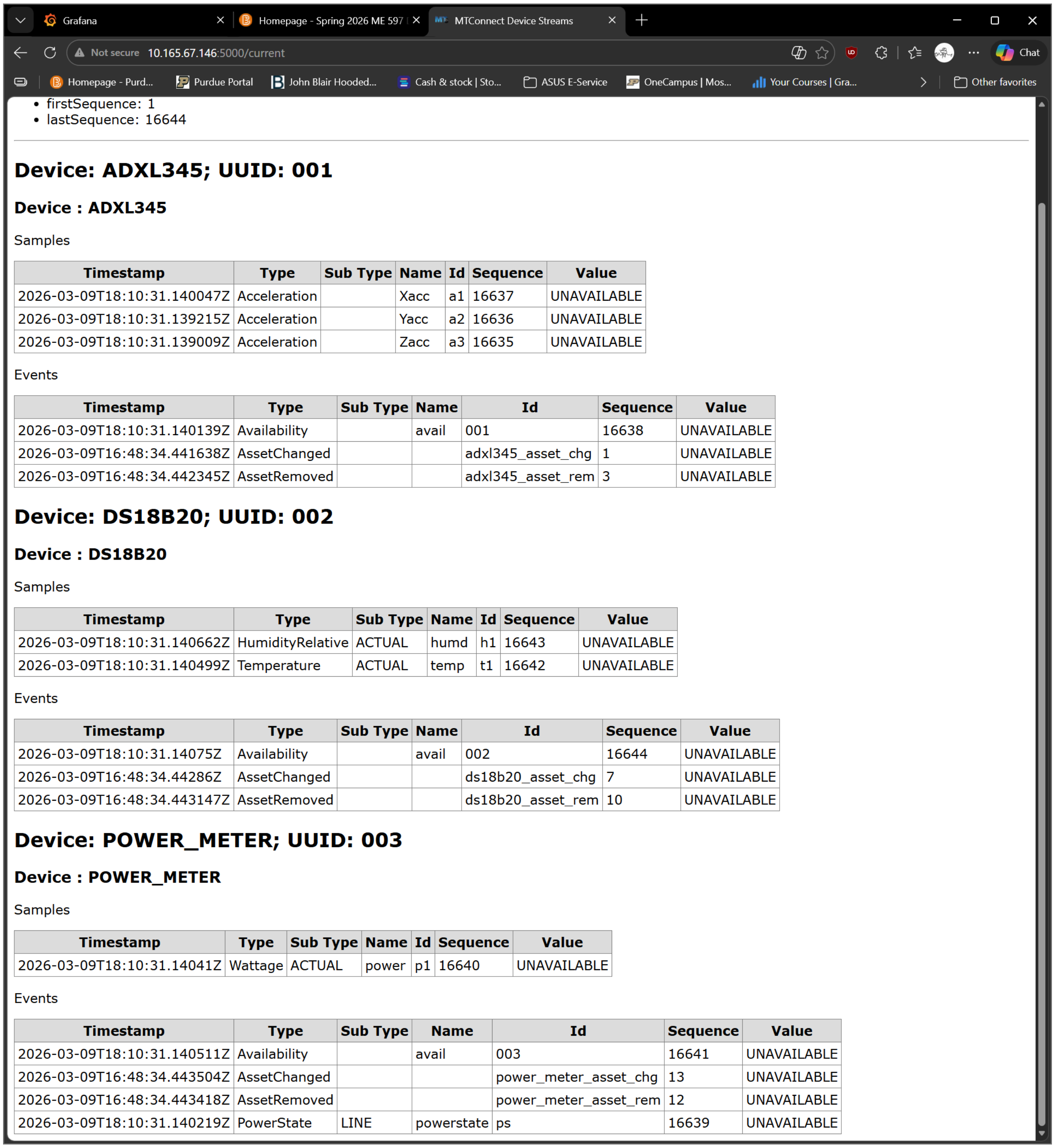


Figure 6 Result grid of grafana_sample table

Task 1.1

Capture your web browser after requesting 'current' information to the MTConnect agent as Figure 5 and attach it below.



- firstSequence: 1
- lastSequence: 16644

Device: ADXL345; UUID: 001

Device : ADXL345

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-03-09T18:10:31.140047Z	Acceleration		Xacc	a1	16637	UNAVAILABLE
2026-03-09T18:10:31.139215Z	Acceleration		Yacc	a2	16636	UNAVAILABLE
2026-03-09T18:10:31.139009Z	Acceleration		Zacc	a3	16635	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-03-09T18:10:31.140139Z	Availability		avail	001	16638	UNAVAILABLE
2026-03-09T16:48:34.441638Z	AssetChanged			adxl345_asset_chg	1	UNAVAILABLE
2026-03-09T16:48:34.442345Z	AssetRemoved			adxl345_asset_rem	3	UNAVAILABLE

Device: DS18B20; UUID: 002

Device : DS18B20

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-03-09T18:10:31.140662Z	HumidityRelative	ACTUAL	humd	h1	16643	UNAVAILABLE
2026-03-09T18:10:31.140499Z	Temperature	ACTUAL	temp	t1	16642	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-03-09T18:10:31.14075Z	Availability		avail	002	16644	UNAVAILABLE
2026-03-09T16:48:34.44286Z	AssetChanged			ds18b20_asset_chg	7	UNAVAILABLE
2026-03-09T16:48:34.443147Z	AssetRemoved			ds18b20_asset_rem	10	UNAVAILABLE

Device: POWER_METER; UUID: 003

Device : POWER_METER

Samples

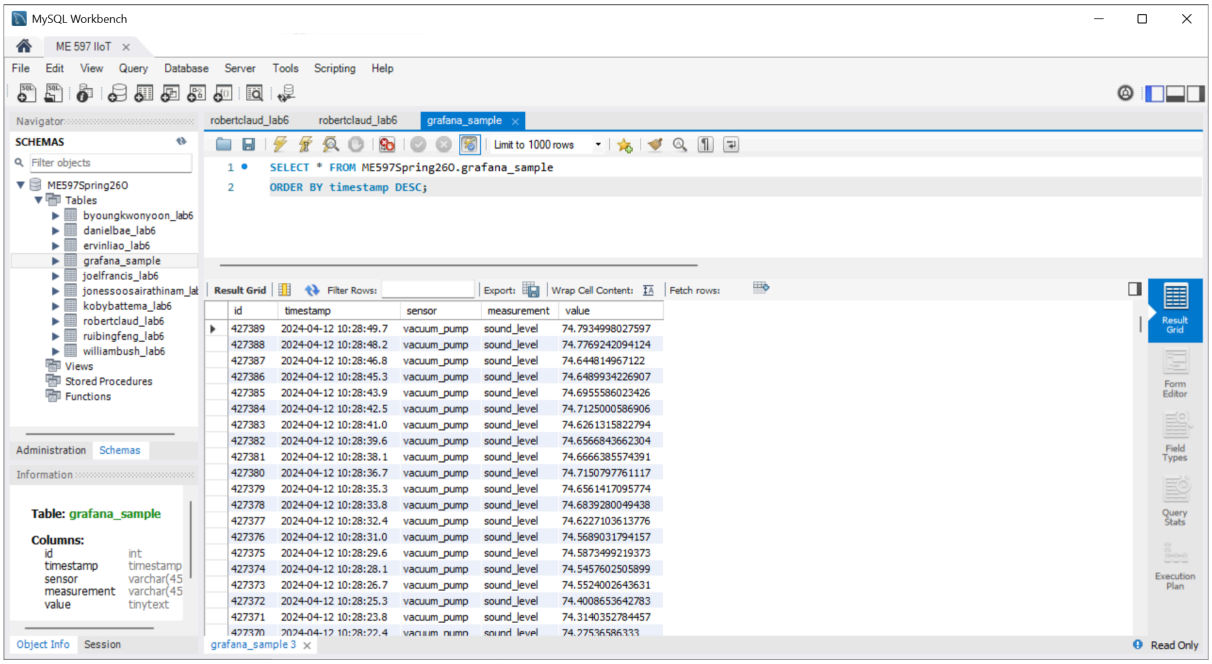
Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-03-09T18:10:31.14041Z	Wattage	ACTUAL	power	p1	16640	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-03-09T18:10:31.140511Z	Availability		avail	003	16641	UNAVAILABLE
2026-03-09T16:48:34.443504Z	AssetChanged			power_meter_asset_chg	13	UNAVAILABLE
2026-03-09T16:48:34.443418Z	AssetRemoved			power_meter_asset_rem	12	UNAVAILABLE
2026-03-09T18:10:31.140219Z	PowerState	LINE	powerstate	ps	16639	UNAVAILABLE

Task 1.2

1. Capture your result grid as Figure 6 after querying the SELECT statement above to ‘grafana_sample’ table and attach it to the report.



2. Answer the questions below by comparing the MTConnect agent and the MysQL database table.

What information in MTConnect agent is stored in the 'timestamp' column?

The information stored in the 'timestamp' column of the MTConnect Agent is the Year, Month, Day, Hour, Minute, and Second data to the precision of 6 decimal places.

What information in MTConnect agent is stored in the 'sensor' column?

The 'sensor' column of the MySQL database table corresponds to the "Device" name in the MTConnect Agent.

What information in MTConnect agent is stored in the 'measurement' column?

The 'measurement' column corresponds to the "Name" column under each MTConnect agent.

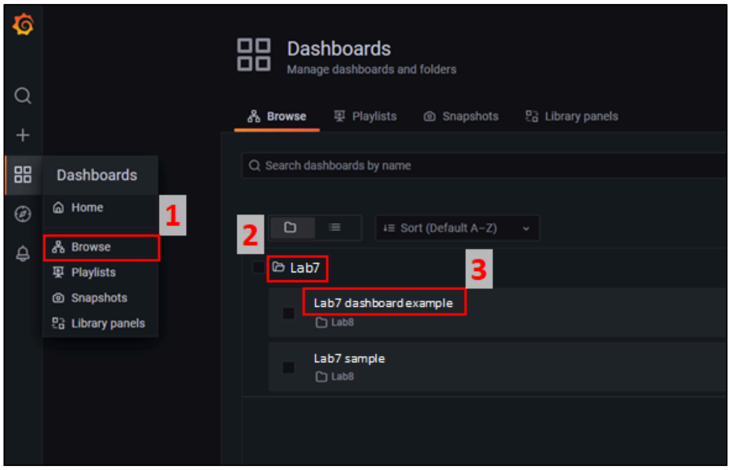
What information in MTConnect agent is stored in the 'value' column?

The 'value' column stores the same information as the "Value" section under each device in the MTConnect agent.

Let’s go back to the Grafana webpage to see the example dashboard. To see the example dashboard (‘Lab7 dashboard example’) page, follow the steps below.

1. Click ‘Browse’ on the ‘Dashboards’ dropdown menu

a. You can see the dropdown menu when you put your mouse pointer on the ‘Dashboards’ icon on the ‘side menu bar’. 2. Click ‘Lab7’ folder 3. Click ‘Lab7 dashboard example’



You will see the example dashboard webpage as Figure 7. The brief description of each data visualization is as follows.

Figure 7 Grafana Lab7 dashboard example

a. ADXL345 data (Sample) as Table

b. Power data (Sample) as Time series

c. Power state data (Event) as Discrete

d. DHT11 temperature data (Sample) as Time series

e. DHT11 humidity data (Sample) as Time series

f. ADXL345 data (Sample) as Time series

As you can see, the default time range of the dashboard is set as last 1 hour. To change the time range, click the time range icon on the right side of the top menu bar and then you will see the time range menu as Figure 8. If you click Last 5 minutes, the time range is changed from the last 5 minutes to now as Figure 9. Also, if you click the refresh icon, you will see the updated dashboard in real time.

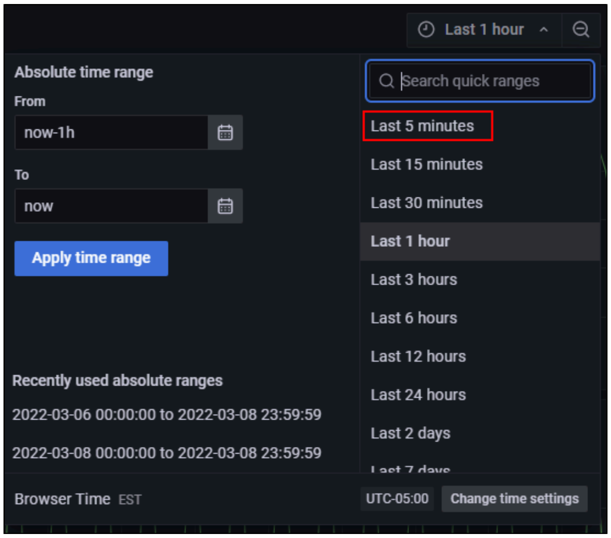


Figure 8 Time range menu of Grafana dashboard

Figure 9 Grafana Lab7 dashboard example after changing the time range to Last 5 minutes

If you want to see a graph or table in detail, click 'View' on the menu on the title of a graph as Figure 10. The example of DHT11 Temperature graph is shown in Figure 11. To go back to the dashboard page, click 'Go back' icon (a back arrow) or hit 'Esc' on your keyboard.

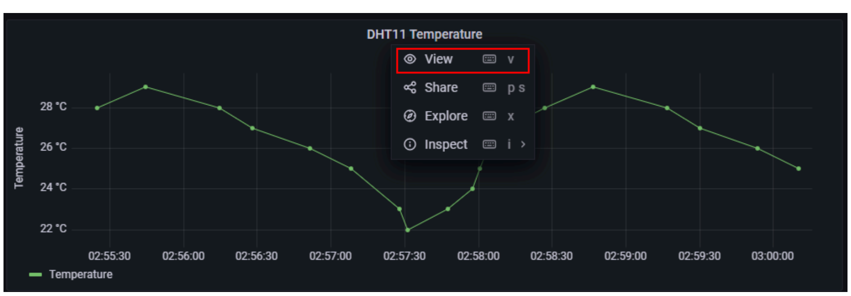


Figure 10 View to see a graph in detail

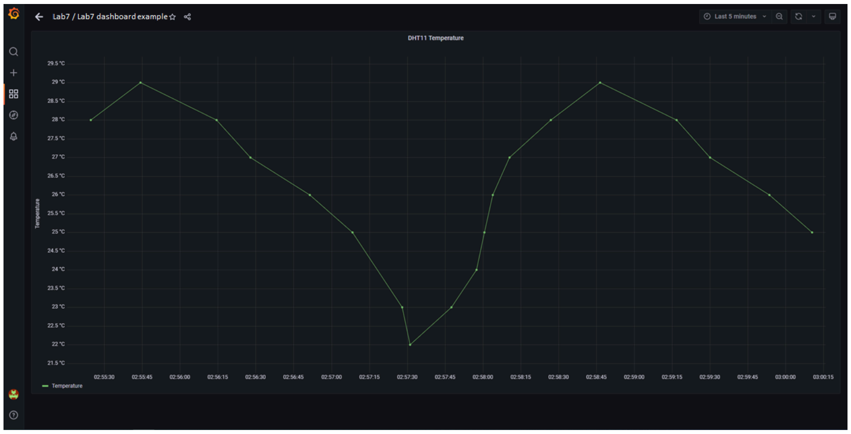
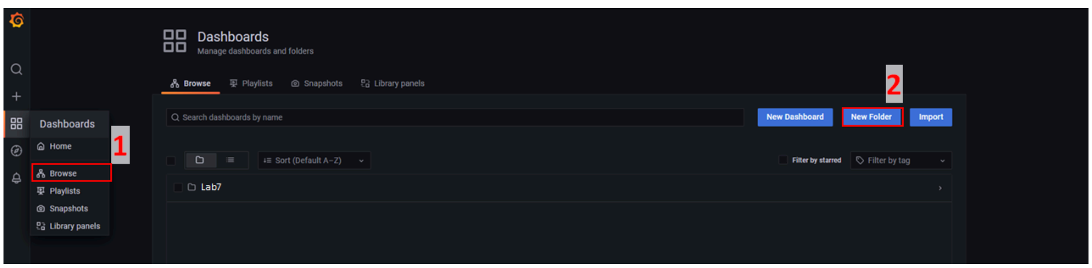


Figure 11 Detail view of DHT11 Temperature graph.

1.4 Create Folder and Dashboard

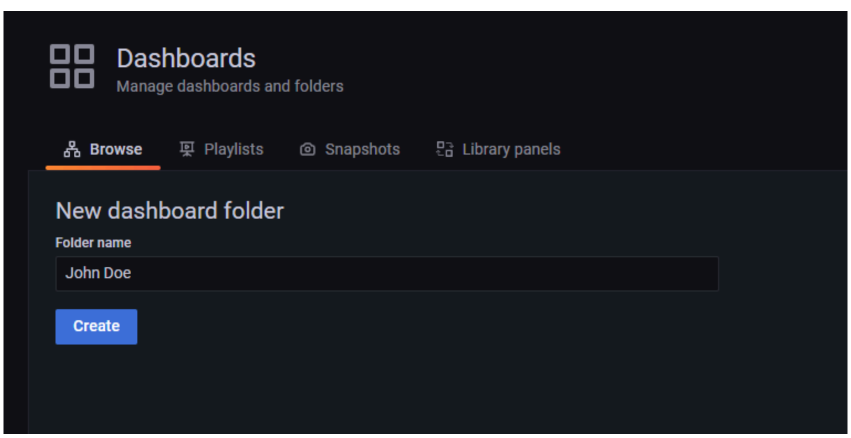
Now, let's create sample dashboards to practice Grafana and SQL query. To create your dashboard page, follow the steps below. Please note that the data source (MySQL database) setup was done by TA.

1. Go back to the 'Dashboards' page by clicking on 'Browse'.
2. Create your folder by clicking on 'New Folder'

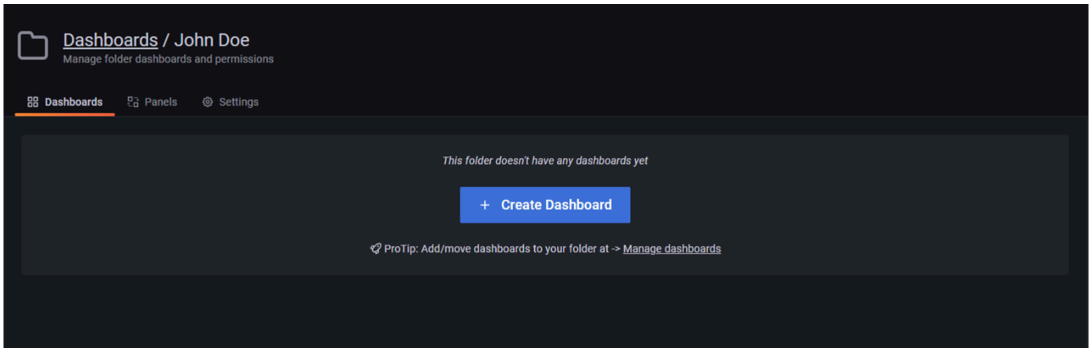


3. Make the folder name as **your name** in the form of 'Firstname Lastname', e.g., John Doe, and then click 'Create'

a. You will work in this folder for the following activities.



4. Click 'Create Dashboard'



You will see the 'New dashboard' as Figure 12. Now, you are ready to create panels. Save the created dashboard as 'Practice' as Figure 13 by clicking the 'Save dashboard' icon on the right side of the top menu bar. Please be make sure that the Folder should be the folder you created.

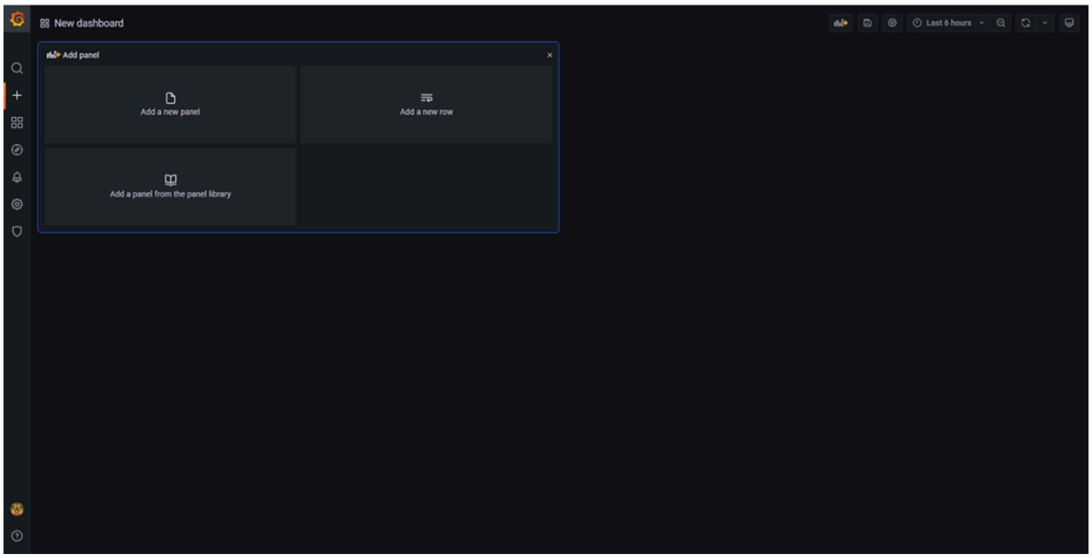


Figure 12 Grafana new dashboard

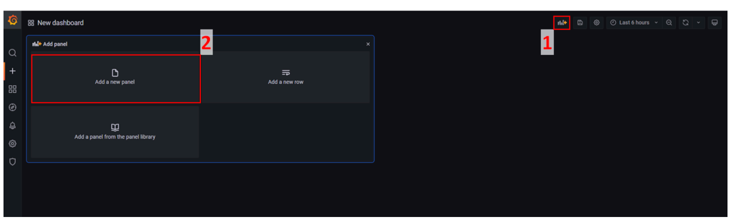


Figure 13 Save dashboard

Please continue to Lab 7.2 here.